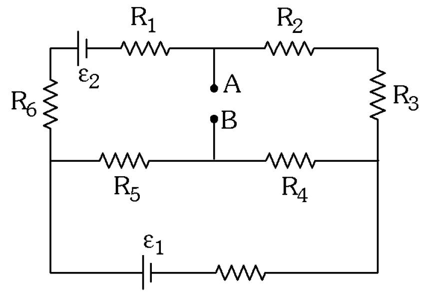
1. In the given circuit diagram, if ideal ammeter connected between point A and B in reading is 5A, if ammeter of resistance 3Ω is connected between A and B its reading is 3A. What will be reading of ideal voltmeter if it is connected between A and B ?

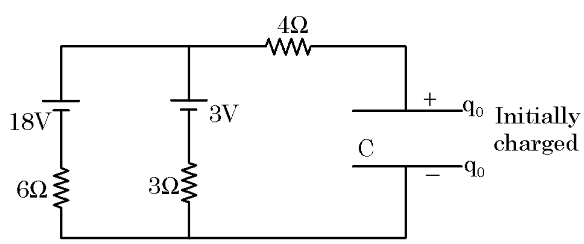


(A) 12.5 volt (B) 10 volt (C) 22.5 volt (D) 31.5 volt

2. A meteorite approaching a planet of mass M (in the straight line passing through the centre of the planet) collides with an automatic space station orbiting the planet in a circular trajectory of radius R. The mass of the station is ten times as large as the mass of the meteorite. As a result of the collision, the meteorite sticks with the station which goes over to a new orbit with the minimum distance R/2 from the planet. Speed of the meteorite just before it collides with the space station is :

(A) $\sqrt{\dfrac{58GM}{R}}$ (B) $\sqrt{\dfrac{38GM}{R}}$ (C) $\sqrt{\dfrac{28GM}{R}}$ (D) $\sqrt{\dfrac{18GM}{R}}$

3. The circuit is completed at t = 0, then total work done by batteries till the capacitor gets full charged.
[q_0 = 64μCand C = 16μF]



(A) 1024 μJ (B) 512 μJ (C) 320 μJ (D) 64 μJ

4. A particle having mass m and charge (–q) moves along an ellipse around a fixed charge Q such that its maximum and minimum distances from the fixed charge are r1 and r2 respectively. The angular momentum L of this particle is :

(A) $\sqrt{\dfrac{mr_1r_2Qq}{2\pi\varepsilon_0(r_1+r_2)}}$ (B) $\sqrt{\dfrac{mr_1r_2Qq}{2\pi^2\varepsilon_0(r_1+r_2)}}$ (C) $\dfrac{1}{2\pi}\sqrt{\dfrac{mr_1r_2Qq}{\varepsilon_0(r_1+r_2)}}$ (D) $\sqrt{\dfrac{m(r_1+r_2)Qq}{2\pi\varepsilon_0}}$

5. C is tensional constant of the wirs supporting the cylinder and θ is the maximum angle rotated by cylinder after magnetisation, value of g is :

(A) $\frac{2mVJ}{e\theta}\sqrt{\frac{1}{Cl_z}}$ (B) $\frac{mVJ}{e\theta}\sqrt{\frac{1}{Cl_z}}$ (C) $\frac{e\theta}{mVJ}\sqrt{\frac{1}{Cl_z}}$ (D) $\frac{e\theta}{2mVJ}\sqrt{\frac{1}{Cl_z}}$

6. A string of linear mass density 5.0×10^{-3} kg/m is stretched under a tension of 65 N between two rigid supports 60 cm apart. If the string is vibrating in its second overtone so that the amplitude at one of its antinodes is 0.25 cm, Choose the correct statement(s) :

(A) The maximum transverse speed of the string at antinodes is (4 ± 0.5) m/s

(B) The maximum acceleration of the string at antinodes is (8000 ± 0.5) m/s^2

(C) The maximum transverse speed of string at a distance of 5 cm from a node is (3 ± 0.5) m/s

(D) The maximum acceleration of string at a distance of 5 cm from a node is (5800 ± 100) m/s

7. The capacitance of a parallel plate capacitor is C_0 when the plates has air between them. This region is now filled with a dielectric slab of dielectric constant K. The capacitor is connected to a cell of emf E and the slab is taken out :

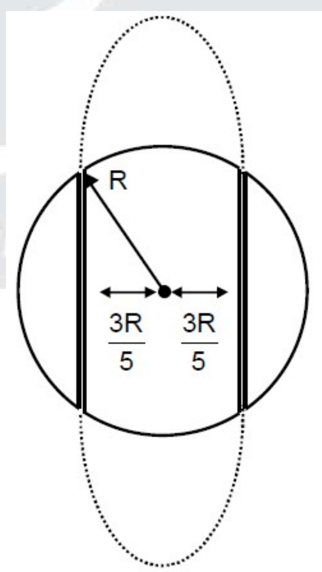
(A) charge $EC_0(K - 1)$ flows through the cell

(B) energy $E^2C_0(K - 1)$ is absorbed by the cell

(C) the energy stored in the capacitor is reduced by $E^2C_0(K - 1)$

(D) the external agent has to do $\frac{1}{2}$ $E^2C_0(K - 1)$ amount of work to take out the slab

8. Consider a uniform spherical planet of mass M and radius R. Two parallel tunnels are dug at perpendicular distance $\frac{3R}{5}$ symmetrically from centre. A particle is to be thrown from the starting of one tunnel such that it enters in another tunnel without making collision with tunnel wall and continues its motion as shown in figure. Tunnels are friction less and particle just fits inside the tunnel. Consider the planet to be at rest.



Choose the correct options.

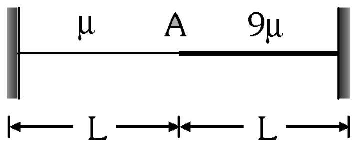
(A) Speed with which the particle is to be thrown at starting of one tunnel is $\sqrt{\frac{GM}{R}}$

(B) Minimum speed of particle during entire motion is $\sqrt{\frac{GM}{9R}}$

(C) Maximum height achieved by the particle during subsequent motion is $\frac{4R}{5}$

(D) Speed with which the particle is to be thrown at starting of one tunnel is $\sqrt{\frac{GM}{9R}}$

9. A light string is tied at one end to a fixed support and to a heavy string of equal length L at the other end as shown in figure. Mass per unit length of the strings are μ and 9μ and the tension is T. Choose the correct statement(s) :



(A) The one of possible values of natural frequencies such that point A is a node is $\frac{3}{2L}\sqrt{\frac{T}{\mu}}$

(B) The one of possible values of natural frequencies such that point A is a node is $\frac{5}{L}\sqrt{\frac{T}{\mu}}$

(C) The one of possible values of natural frequencies such that point A is a antinode is $\frac{3}{4L}\sqrt{\frac{T}{\mu}}$

(D) The one of possible values of natural frequencies such that point A is a antinode is $\frac{7}{4L}\sqrt{\frac{T}{\mu}}$

10. A point charge of specific charge $\frac{q}{m}$ = 0.1 C/kg is projected in uniform magnetic field. The particle moves in magnetic field such that its position vector at any instant is given by $\vec{r} = 3\sin t\hat{i} + 3\cos t\hat{j} + 4t\hat{k}$. Select correct statements from following :

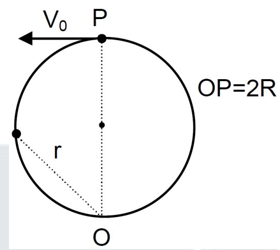
(A) Magnetic field in space is 10T

(B) The distance traveled by the particle in 5s is 20m

(C) Power of magnetic force is zero

(D) The radius of curvature of the path is 3m

11. In a certain universe, the gravity is a conservative force but not necessarily an inverse square law force. In that universe, there exists a single point like mass 'm' that is influenced only by gravity. As a result of a gravitational force, the mass orbits the fixed source (point O) in a circle, so that the source point is on the circle's rim. Neglecting the singular point, select that correct alternative/s. ('r' is the distance of 'm' from O)



(A) Velocity of the point mass as function of r is given by $\frac{4R^2V_0}{r^2}(r \neq 0)$

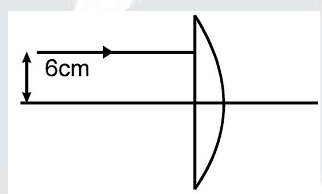
(B) Force experienced by point mass depends on initial speed V_0 of point mass.

(C) Force experienced by point mass will be inversely proportnal to fifth power of its distance from O.

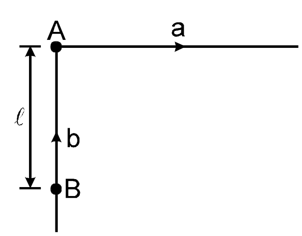
(D) Position vector of particle with respect to centre of circle sweeps out equal area in equal interval of time.

12. Four point charge q, – q, 2Q and Q are placed in order at the corners A, B, C, D of a square. If the field at the mid point of CD is zero, then the value of q/Q is $\frac{5\sqrt{5}}{x}$. Find the value of x

13. A light ray parallel to the principal axis is incident (as shown in the figure) on a planoconvex lens with radius of curvature of its curved part equal to 10 cm. Assuming that the refractive index of the material of the lens is 4/3 and medium on both sides of the lens is air, the distance of the point from the curved surface of lens where this ray meets the principal axis is $\frac{26y}{7}$ cm then find value of y.

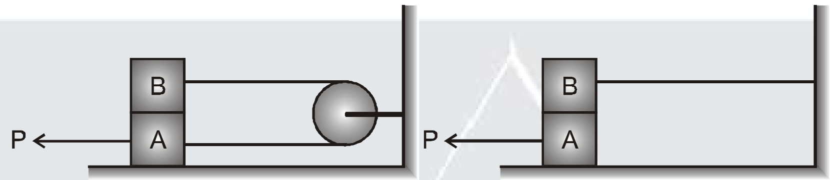


14. In the figure shown, A and B are two particles which start from rest. A has constant acceleration a=3 m/s^2 in the direction shown. B also increases its speed at a constant rate b = 4m/s^2 , but the direction of velocity is always towards A. If the time after which B meets A is $\sqrt{\frac{2\ell b}{b^2+\alpha}}$ then, α (in SI unit) is

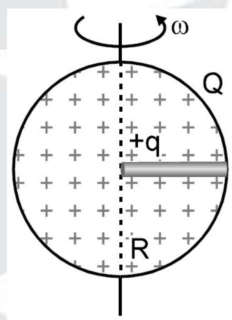


15. Consider a uniformly charged non-conducting cube of volume charge density ρ. Potential of one of vertex is 4 volt, find the potential of centre of the cube (in volts).

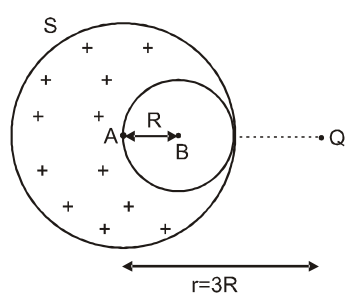
16. Each of the two block shown in the figure has mass m. The pulley is smooth and the coefficient of friction for all surfaces in contact is μ. A constant horizontal force P applied in two cases shown in such a way that block A start just sliding. If the ratio of minimum force P in case-I and case-II is $\frac{m}{n}$. If m and n are integers and $\frac{m}{n}$ is in its lowest form, then value of m + n is.



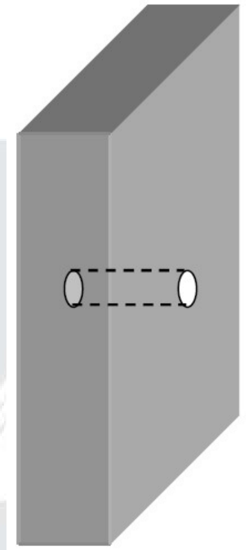
17. A small ball of mass 1 kg and charge $\frac{2}{3}\mu C$ is placed at the centre of a uniformly charged sphere of radius 1 m and charge $\frac{1}{3}$ mC. A narrow smooth groove is made in the sphere from centre to surface as shown in figure. The sphere is made to rotate about a diameter perpendicular to the groove at a constant rate of $\frac{1}{2\pi}$ revolutions per second. Find the speed w.r.t. ground (in m/s) with which the ball slides out from the groove. Neglect any magnetic effect.



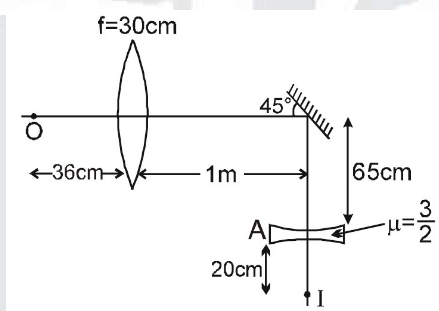
18. A point charge Q is kept at distance r = 3R from the centre (point A) of a uniformly charged sphere having volume charge density ρ and radius 2R. The sphere has a spherical cavity of radius R having centre at point B such that AB = R. If the interaction energy between the point charge and the sphere having cavity is given as $\frac{13Q\rho R^2}{N6\varepsilon_0}$ .Find N ? (Line joining points A and B passes through point charge Q)



19. Consider an asteroid in the shape of a slab with thickness D, with the uniform volume mass density ρ. Assume that the two dimensions perpendicular to the thickness D are essentially infinite (much larger than D). A thin tube is drilled along the "D" direction, as shown in figure. If an object is dropped from rest into the tube from one end, the angular frequency of its oscillatory motion is $\sqrt{N\pi G\rho}$. Find the value of N



20. The final image I of the object O shown in the figure is formed at point 20 cm below a thin equi-concave lens, which is at a depth of 65 cm from principal axis. From the given geometry, calculate the radius of curvature in cm of lens kept at "A". (Refractive index of equi-convex lens is 1.5 and placed in air).



ANSWER KEY

1.	(C)	**2.**	(A)	**3.**	(B)	**4.**	(A)	**5.**	(A)
6.	(ABCD)	**7.**	(ABD)	**8.**	(ABC)	**9.**	(ACD)	**10.**	(AC)
11.	(ABC)	**12.**	2	**13.**	5	**14.**	–9	**15.**	8
16.	+7	**17.**	2 m/s	**18.**	3	**19.**	4	**20.**	60